

# S-BFP Artifact Appendix

From Frozen Pose to Live Dance: Stochastic Browser Fingerprinting for Robust Risk-Based Authentication

## Artifact overview

S-BFP is a Flask and browser research prototype that turns a static browser fingerprint into a server-controlled challenge-response measurement. A deterministic random bit generator derives device-specific primitives, while a session input changes their spatial arrangement. The artifact implements Canvas text and Web Audio waveforms.

The package supports (1) functional inspection of both browser workflows and (2) independent recomputation of the retained stability rows and environment table from a de-identified 213-participant summary dataset.

## Claims and evaluation mapping

| Paper claim         | Evaluation action             | Expected result                       |
|---------------------|-------------------------------|---------------------------------------|
| Audio stability     | Run reproduction script       | 206/206 (100.0%)                      |
| Canvas stability    | Run reproduction script       | 196/198 (99.0%)                       |
| Cohort distribution | Inspect generated Table 3 CSV | 213 devices; exact row match          |
| Executable workflow | Run both browser panels       | Repeated render and stability summary |

## Scope

The artifact does not implement a complete production risk engine and does not claim to prove security against every adaptive attacker. The restricted per-participant source collection is not part of the release package.

## Requirements and installation

Hardware: any x86-64 or ARM64 CPU, 1 GiB free memory, and about 300 MiB disk for native use. A discrete GPU is not required. Software: Python 3.11-3.13 and a current browser with Canvas and Web Audio. Docker with Compose v2 is optional.

### Windows PowerShell

```
.\scripts\setup.ps1  
.\scripts\run.ps1
```

### Linux or macOS

```
sh scripts/setup.sh  
sh scripts/run.sh
```

### Docker alternative

```
docker compose up --build
```

Open <http://127.0.0.1:5001/>. Native startup normally opens this address automatically. The aggregate reproduction is CPU-only and typically finishes in under one minute. Interactive rendering usually takes several seconds per modality.

## Functional evaluation

1. Select Register and enter a pseudonymous username of 3-20 letters, digits, or underscores.
2. Read and affirm the consent notice, then save the generated password.
3. Acquire the exclusive test session.
4. Run Canvas and Audio. Canvas performs five repetitions; Audio performs ten.
5. Confirm that both panels present a completion message. A changed hash is a legitimate instability result.
6. With default settings, the new runtime record must not contain an IP address, full user-agent string, or raw Canvas fingerprint.

## Paper-table reproduction

```
python scripts/reproduce_tables.py --verify-paper
```

## Expected reproduction result

Expected values: Audio 206/206; Canvas 196/198; environment total 213.

## Automated and package checks

```
python -m unittest discover -s tests -v
python scripts/build_artifact.py
python scripts/verify_artifact.py dist/s-bfp-usenix-artifact.zip
```

The tests use an isolated temporary data directory. The package verifier requires the documentation and public dataset, verifies SHA-256 entries, checks the 213-row schema, and rejects restricted source, runtime-data, upload, cache, and Git paths.

## Released data

data/public/records.jsonl is a deterministic aggregate of each participant's first collection session, matching the paper's cohort snapshot. Each row contains only a new sequential ID, coarse operating-system and browser categories, and modality availability, run count, unique-result count, and stability boolean.

Original usernames, password material, IP addresses, full user agents, timestamps, renderer strings, seeds, rendering hashes, raw images, and audio samples are omitted. Evaluators do not need the restricted source records.

## Limitations and troubleshooting

If port 5001 is occupied, set S\_BFP\_PORT. Docker reproduces the server environment, not the host browser audio or font stack. The process-local session coordinator supports one active evaluator and is not a multi-worker production design.

## Submission notes

For an Available badge, deposit the exact sanitized release on a permanent platform and replace repository placeholders with the immutable version DOI. Before public release, authors must select explicit code/data licenses, add camera-ready citation metadata, reconcile the ethics statement with retained source fields, and verify a fresh extraction on the target platforms.